

Near Real-Time rPPG Integration

Build a small prototype that takes a 60-second face video as input, processes it in 5-second chunks, and generates:

1. BPM estimate per 5-second chunk
2. Overall BPM estimate for the full 60 seconds
3. Basic runtime/performance metrics

You may choose the stack and implementation approach. We care most about your ability to integrate a CV model into a near real-time pipeline and about the accuracy of the data. You can use any of the the open-source rPPG models listed below (or any other) as the starting point:

- <https://github.com/KegangWangCCNU/open-rppg>: Heart Rate and Respiratory Rate
- <https://github.com/ubicomplab/rPPG-Toolbox>: Heart Rate and Respiratory Rate
- <https://github.com/prouast/heartbeat>: Heart Rate
- <https://github.com/prouast/heartbeat>: Heart Rate
- <https://github.com/eugenelet/Meta-rPPG>: Heart Rate

Special bonus point for including Respiratory Rate and any other biomarkers as part of the integration workflow.

Expected output:

- Link to test prototype
- Sample output showing chunk-level BPM + final BPM
- Any notes on model performance, latency, and failure cases
- We strongly expect candidates to use AI as much as possible. We only ask that you share how you used AI tools.

Evaluation criteria:

- Performance of the model (accuracy of the estimation)
- Clean integration of the rPPG model
- Ability to process video incrementally in 5-second windows
- Reasonable BPM aggregation logic
- Code clarity and reliability
- Practical thinking around real-time deployment constraints